INCH-POUND
MIL-PRF-19500/782
24 July 2024

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, GALLIUM NITRIDE, HIGH ELECTRON MOBILITY TRANSISTOR (HEMT),
RADIATION HARDENED, ENHANCEMENT MODE,
TYPES 2N7667UFB, 2N7668UFB, 2N7669UFB, QUALITY LEVELS JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of
this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an enhancement-mode Gallium Nitride, radiation hardened (Total Dose and Single Event Effects (SEE)), High Electron Mobility Transistor (HEMT). Two levels of product assurance (JANTXV, and JANS) are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type. Enhancement Mode Gallium Nitride devices are not sensitive to total ionizing dose and will be provided as level H for JANTXV and JANS product assurance levels.

1.2 Package outlines. The device package for the encapsulated device types are as follows: (UFBC and UFBM) in accordance with figure 1. The dimensions and topography for JANHC and JANKC unencapsulated die is as follows: The A version die in accordance with figure 2 and figure 3.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type (1)	P_T T_C = +25°C (2)	P_T T_A = +25°C (free Air)	$R_{\theta JC}$ (3)	V_{DS}	V_{GS}	I_{D1} T_C = +25°C (4)	I_{D2} T_C = +100°C (4)	I_D Pulse	T_J and T_{STG}
	W	W	°C/W	V dc	V dc	A dc	A dc	A (pk)	°C
2N7667UFBC	56	2.78	2.25	40	-4, +6	30	19	120	-55
2N7668UFBC	56	2.78	2.25	100	-4, +6	30	18	120	to
2N7669UFBC	31	2.23	4.02	200	-4, +6	18	11	72	+150

(1) Also applies to UFBM suffix versions.
(2) Derate linearly by 0.45W/°C for T_c > +25°C for 2N7667 and 2N7668 and 0.25W/°C for 2N7669.
(3) See figure 4, thermal impedance curves.
(4) The following formula derives the maximum theoretical I_D limit: I_D may be limited by package

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type (1)	V_{DS} rated	$V_{GS\ (th)1}$	Max I_{DSS1}	Max $r_{DS(on)}$ (2) V_{GS} = 5.0 V dc I_{D1}		Max DR_{DSON} V_{GS} =5.0 V (3)
		V_{DS} = V_{GS} I_D = 5 mA dc	V_{GS} = 0 V V_{DS} = 100% rated V_{DS}	T_J = +25°C	T_J = +125°C	T_J = +25°C $V_{DS(off)}$ = V_{DS} rated
	V dc	V dc Min	Max	µA dc	Ohm	Ohm	Ohm
2N7667UFBC	40	0.8	2.5	400	0.01	0.018	0.013
2N7668UFBC	100	0.8	2.5	250	0.014	0.025	0.018
2N7669UFBC	200	0.8	2.5	250	0.030	0.050	0.039

(1) Also applies to UFBM suffix versions.
(2) Pulsed (see 4.5.1).
(3) According to JEP173 within 10us.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JANTXV” and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "H".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "7667” and “7668", and “7669”.

1.5.4.3 Suffix letters. The following suffix letters are incorporated in the PIN for this specification sheet:

UFBC	Indicates a 4 pad surface mount package with ceramic lid.
UFBM	Indicates a 4 pad surface mount package with metal lid.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A".

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1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N7667UFBC, 2N7668UFBC, 2N7669UFBC: ..... 1 Mrad(Si) 1/
For device type 2N7667UFBM, 2N7668UFBM, 2N7669UFBM: .....1 Mrad(Si) 1/

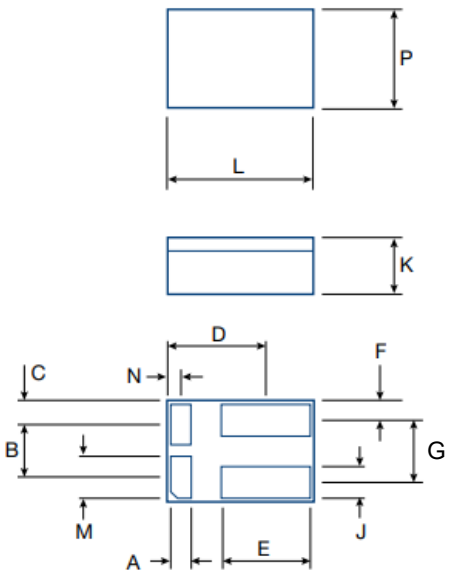
1.6.2 Heavy ion irradiation SEE. SEE testing includes Single-Event Burnout (SEB), Single-Event Dielectric Rupture (SEDR), and Single-Event Gate Rupture (SEGR) tests at the specified linear energy transfer (LET) units. See 6.7 for the specific RHA SEE characterization testing details. The following characterization conditions were used for the testing to the requirements herein:

For device types 2N7667UFBC, 2N7668UFBC, 2N7669UFBC:
No SEB, SEDR, and SEGR were observed at surface LET (see table IV) .....≤ 84 MeV·cm2/mg (Si) 2/
(In-situ bias conditions: VDS = 40 V and VGS = -4 V)
(In-situ bias conditions: VDS = 100 V and VGS = -4 V
(In-situ bias conditions: VDS = 150 V and VGS = -4 V)

1/ Manufacturer supplying these device types have performed characterization testing in accordance with MIL-STD-750, method 1019, condition A (dose rate = 50 - 300 rad(Si)/s). The radiation end point limits are guaranteed only for the conditions as specified in MIL-STD-750, method 1019, condition A to a maximum total ionizing dose level of 1 Mrad(Si). The test is performed per wafer lot on a 10 devices per bias.

2/ Manufacturer also performed heavy ion SEB, SEDR, and SEGR test at TAMU Radiation Effects Facility in accordance with TM1080 of MIL-STD-750. Limits are characterized at initial qualification and after any design or process changes which may affect the SEE (SEB/SEDR/SEGR) characteristics. This test is performed on every wafer fabrication lot. For more information on SEE (SEB/SEDR/SEGR) test results, customers are requested to contact the manufacturer.

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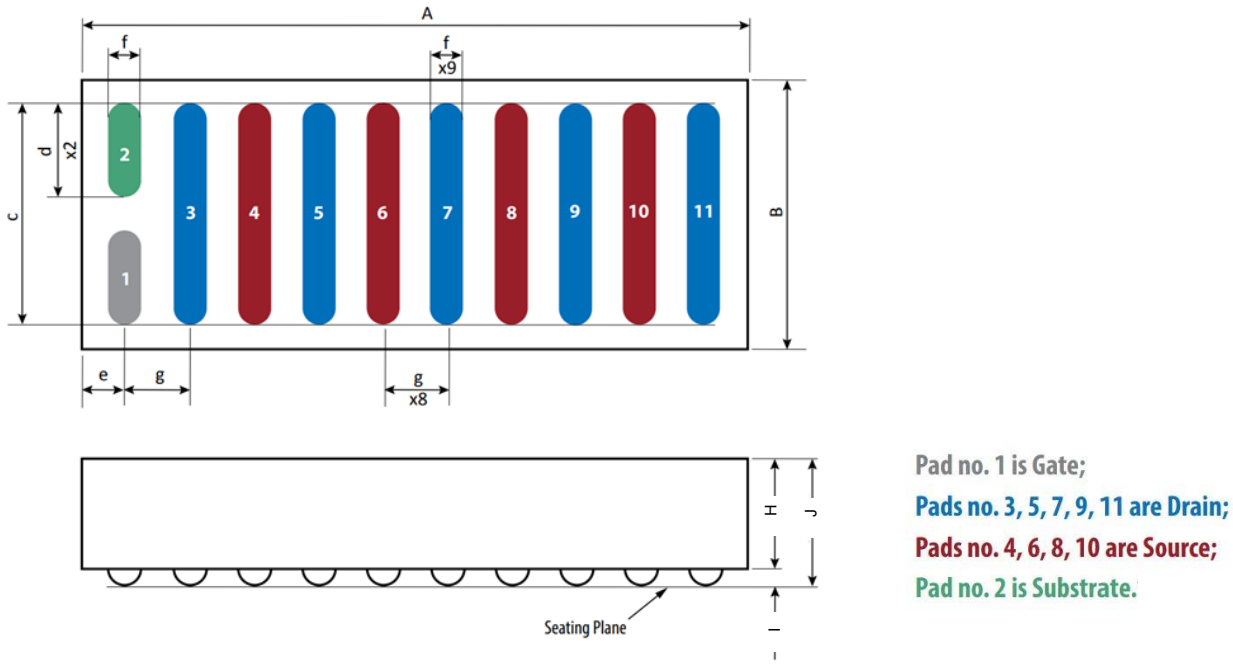
Ltr	Dimensions			
	In	In	mm	mm
	Min	Max	Min	Max
A	.027	.037	0.685	0.939
B	.073	.083	1.854	2.108
C	.031	.041	0.784	1.041
D	.143	.153	3.632	3.886
E	.129	.139	3.277	3.531
F	.027	.037	0.686	0.940
G	.082	.092	2.083	2.337
J	.050	.060	1.270	1.524
K	.078	.088	1.981	2.235
L	.215	.225	5.461	5.715
M	.058	.068	1.473	1.727
N	.016	.026	0.406	0.660
P	.145	.155	3.683	3.937

UFB Package

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.

FIGURE 1. Physical dimensions, surface mount UFBC and UFBM.

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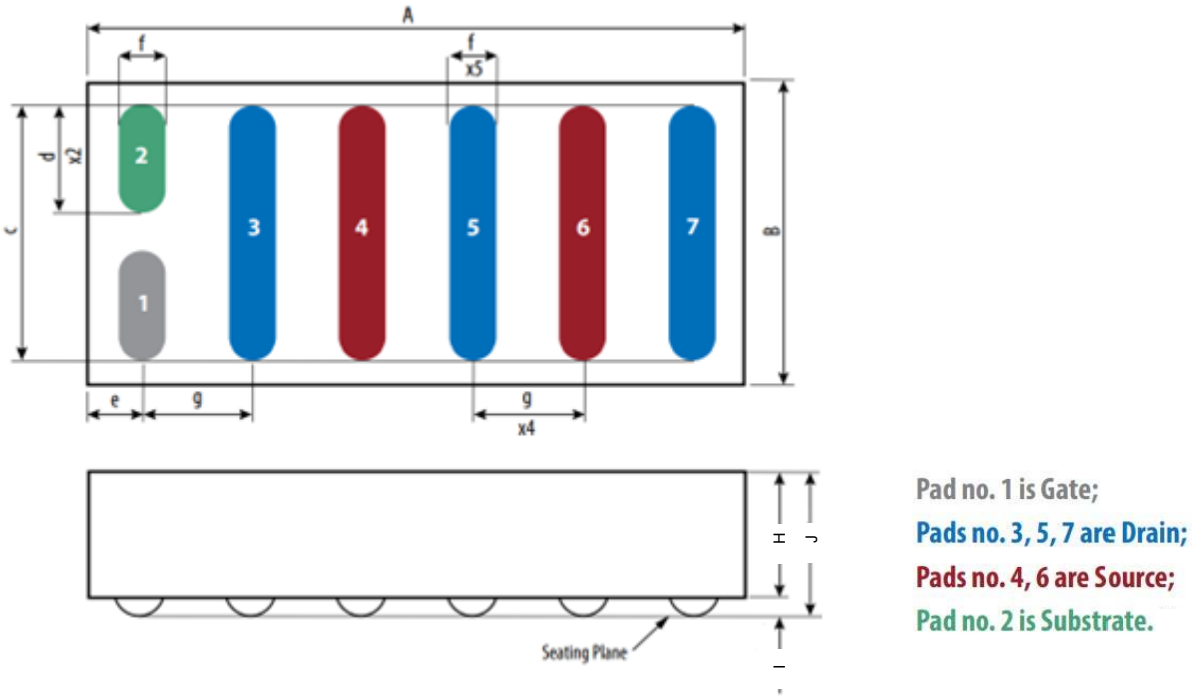
Ltr	Dimensions					
	In	In	In	µm	µm	µm
	Min	Nom	Max	Min	Nom	Max
A	.1604	.1616	.1628	4075	4105	4135
B	.0631	.0644	.0654	1602	1635	1662
c	.0543	.0544	.0545	1379	1382	1385
d	.0227	.0228	.0230	577	580	583
e	.0093	.0098	.0104	235	250	265
f	.0077	.0079	.0081	195	200	205
g	.0157	.0157	.0157	400	400	400
H		.027			685	
I	.0031	.0039	.0047	80	100	120
J			.0321			815

NOTES:

1. Top metal: 87µm nominal 95Pb5Sn bump on top of 4µm nominal of Aluminum.
2. Back metal: N/A
3. Substrate pad should be connected to source.

FIGURE 2. Physical dimensions JANHCA and JANKCA 2N7667 and 2N7668 version die dimensions.

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Ltr	Dimensions					
	In	In	In	µm	µm	µm
	Min	Nom	Max	Min	Nom	Max
A	.1387	.1399	.1411	3523	3553	3584
B	.0630	.0643	.0654	1600	1633	1661
c	.0543	.0544	.0545	1379	1382	1384
d	.0227	.0228	.0230	5777	579	584
e	.0103	.0109	.0115	262	277	292
f	.0096	.0098	.0100	244	249	254
g	.0236	.0236	.0236	599	599	599
H		.027			685	
I	.0031	.0039	.0047	80	100	120
J			.0321			815

NOTES:

1. Top metal: 87µm nominal 95Pb5Sn bump on top of 4µm nominal of Aluminum.
2. Back metal: N/A
3. Substrate pad should be connected to source.

FIGURE 3. Physical dimensions JANHCA and JANKCA 2N7669 version die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

JEDEC - SOLID STATE TECHNOLOGY ASSOCIATION (JEDEC)

JEP138 - User Guidelines for IR Thermal Imaging Determination of Die Temperature

JEP173 - Dynamic ON-Resistance Test Method Guidelines for GaN HEMT based Power Conversion Devices.

JEP180 - GUIDELINE FOR SWITCHING RELIABILITY EVALUATION PROCEDURES FOR GALLIUM NITRIDE POWER CONVERSION DEVICES

JEP186 - Guideline to Specify a Transient Off-State Withstand Voltage Robustness Indicator in Datasheets for Lateral GaN Power Conversion Devices

(Copies of these documents are available online at https://www.jedec.org or from JEDEC - Solid State Technology Association, 3103 North 10th Street, Suite 240-S, Arlington VA 22201-2107)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

GaN…….Gallium Nitride
HEMT……..High Electron Mobility Transistor
nC ..... ….nano Coulomb.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Multiple chip construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. Marking on the UF package shall consist of an abbreviated part number, the date code, and the manufacturer's symbol or logo. The prefixes JANTXV and JANS can be abbreviated as JV and JS respectively. The "2N" prefix and the "UF" suffix can also be omitted. The radiation hardened designator H, shall immediately precede (or replace) the device "2N" identifier (depending upon degree of abbreviation required).

3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, in HEMT areas.

f. Maintain relative humidity above 30 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ kΩ, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.1.1 SEE. SEE shall be performed at initial qualification, on every wafer fabrication lot, and after process or design changes which may affect radiation hardness (see table III and table IV). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the slash sheet. End-point measurements shall be in accordance with table II. SEE characterization data shall be made available upon request of the qualifying or acquiring activity.

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4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS	Measurements for JANTXV
4	Method 2006 of MIL-STD-750	
5	Method 2052 of MIL-STD-750	
7	Method 1071 of MIL-STD-750	
9	Subgroup 2 of table I herein I_{DSS1}, I_{GSSF1}, I_{GSSR1} as minimum	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)1}$, $V_{GS(TH)1}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±15 µA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±15 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)1}$, $V_{GS(TH)1}$ Subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±15 µA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±15 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±30 percent of initial value. $\Delta V_{GS(TH)1}$ = ±30 percent of initial value.	Subgroup 2 of table I herein ΔI_{GSSF1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±15 µA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±15 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±30 percent of initial value. $\Delta V_{GS(TH)1}$ = ±30 percent of initial value.
14	Dynamic $R_{DS(ON)}$ JEP173 Double Pulse, see 4.5.2	
15	Method 1071 of MIL-STD-750, a. test condition A b. fine leak	Method 1071 of MIL-STD-750, a. test condition A b. fine leak
16	Method 2076 of MIL-STD-750, See 4.5.3	Method 2076 of MIL-STD-750, See 4.5.3
17	Subgroup 2 of table I herein	
18	Method 2071 of MIL-STD-750	

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles.
B3	2077	SEM.
B4	1042	Intermittent operation life, condition D, t_{on} = 30 seconds minimum.
B5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated V_{GS}; T_A = +150°C, t = 48 hours minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated V_{DS}; T_A = +150°C, t = 240 hours minimum.
B5	JEP173	Dynamic R_{DSON}, V_{GS} =5.0 V, V_{DS}-stress = 80 percent rated, Stress time = 10 s
B5	2037	Not applicable.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition C, 25 cycles.
B3	1042	Intermittent operation life, condition D. t_{on} = 30 seconds minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

Subgroup	Method	Condition
C2	2036	Terminal strength is not applicable.
C6	1042	Intermittent operation life, condition D. t_{on} = 30 seconds minimum.
C6	2037	Not applicable.
C6	2017	Die Shear / Die Pull. 6 devices.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

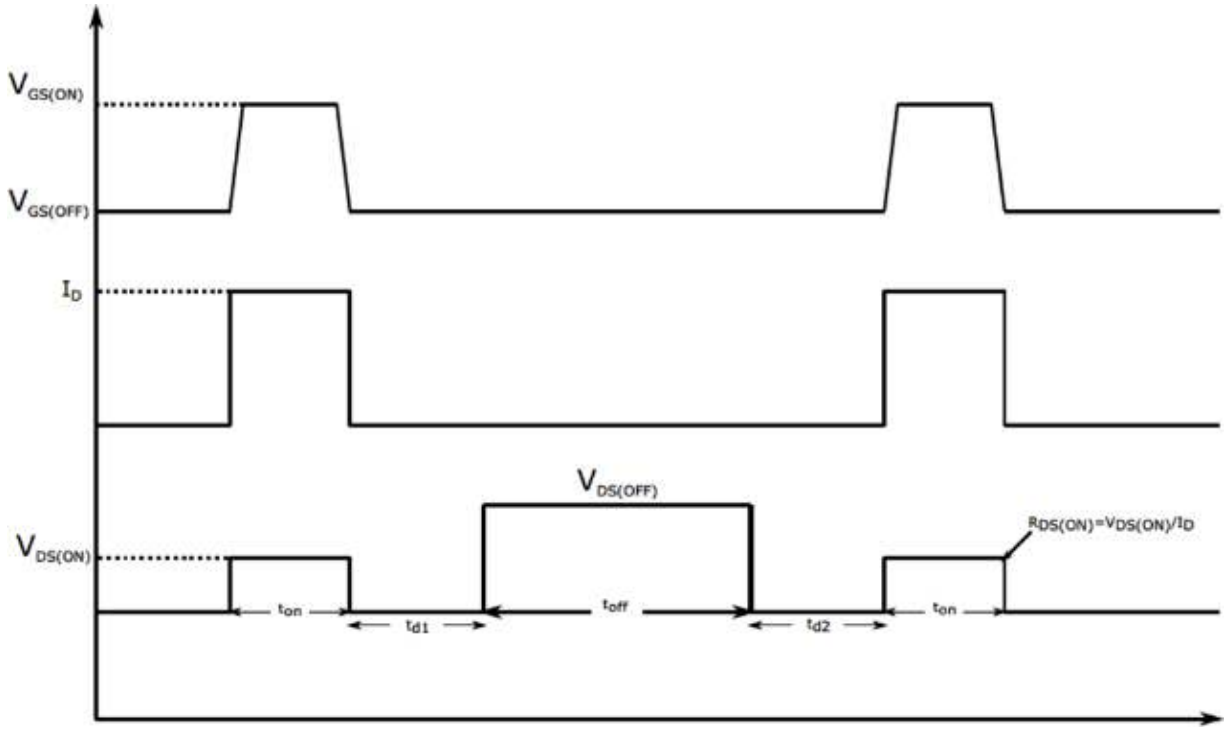
4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Dynamic R_{DSON}. The DR_{DSON} is performed according to the Pulse Current Voltage method in JEP173. V_{GS} = 5.0 V, V_{DS}-stress = 80 percent rated, Stress time = 10 s, t_{on} is determined by tester configuration and should be short enough to not heat the device. t_{d2} should be less than 10µs or as fast as possible to avoid trap annealing.

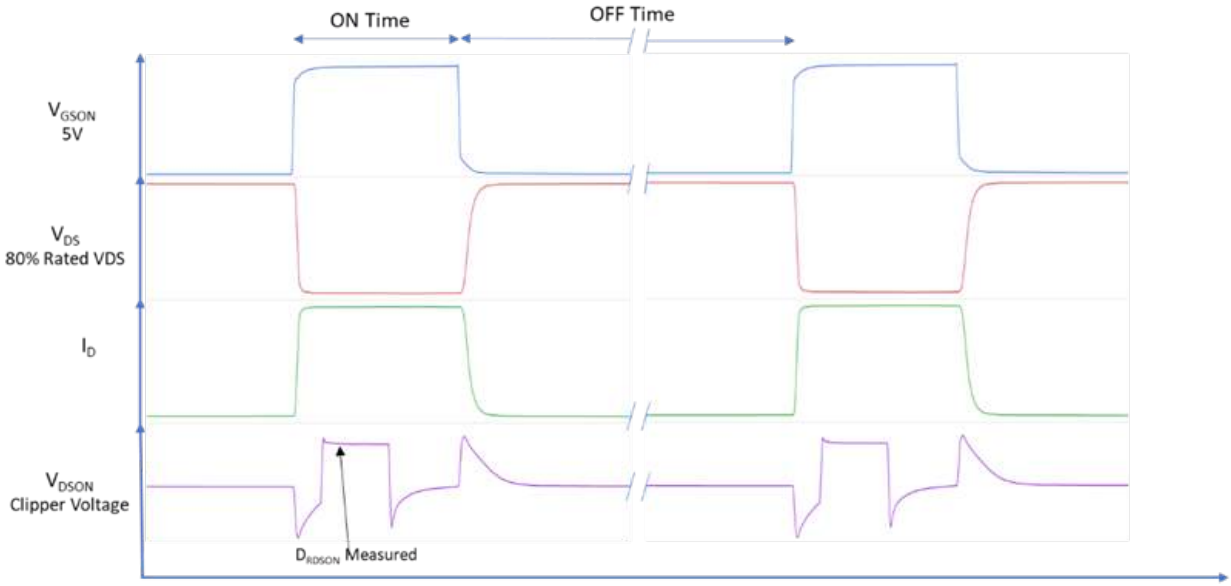


4.5.3 Radiography. Total contact area voids greater than 15% of the total intended contact area is rejectable. If 5% or more of the devices in a JANS inspection lot fail this die attach criteria, the lot shall be reviewed, and appropriate corrective action taken. Any bridging between conductive paths or more than one bridge within a conductive path is rejectable. In addition, if the die attach image shows any unusual anomalies, the lot shall be reviewed, and appropriate corrective action shall be taken.

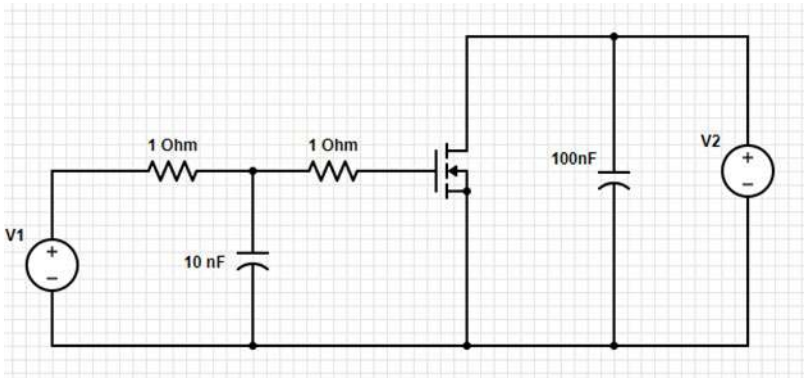
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4.5.4 Switching dynamic RDSon. The switching Dynamic RDS_{ON} is performed within the guidelines of JEP180 in a hard switch test vehicle. The circuit switches a resistive load and monitors RDS_{ON} during the ON time based on the following criteria:

V_{DS} = 80% of rated voltage
I_D = 40% - 60% of rated current set by resistive load to properly measure the on-state voltage drop
T_J = 80°C
Duty Cycle = 0.1 %
Frequency ≥ 500Hz



4.5.5 Single Event Effects: When establishing the test circuit for the SEE testing, the gate resistance should be kept as low as possible (less than or equal to 1 Ohm) as shown in the circuit below. This is to prevent the gate from turning on during SEE transient.



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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750 or industry standard		Symbol	Limits	Limits	Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Gate to source voltage (threshold)	3403	V_{DS} = V_{GS}, I_D = 5 mA dc	$V_{GS(TH)1}$	0.8	2.5	V dc
Gate current	3411	V_{GS} = +6 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		500	µA dc
Gate current	3411	V_{GS} = -4 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}			
2N7667UFBC					-400	µA dc
2N7668UFBC					-250	µA dc
2N7669UFBC					-150	µA dc
Drain current	3413	V_{GS} = 0, bias condition C, V_{DS} = 100 percent of rated V_{DS}	I_{DSS1}			
2N7667UFBC					400	µA dc
2N7668UFBC					250	µA dc
2N7669UFBC					250	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 5 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D1}	$r_{DS(ON)1}$			
2N7667UFBC					0.01	Ω
2N7668UFBC					0.014	Ω
2N7669UFBC					0.028	Ω
Dynamic ON-Resistance 2/	JEP173	See 4.5.2	DR_{DSON}			
2N7667UFBC					0.013	Ω
2N7668UFBC					0.018	Ω
2N7669UFBC					0.039	Ω
Forward voltage	4011	V_{GS} = 0, condition A, pulsed (see 4.5.1), I_D = 0.5A	V_{SD}		3.0	V (pk)
Subgroup 3						
High temperature operation:		T_C = T_J = +125°C				
Gate current	3411	V_{GS} = +6 V dc, bias condition C, V_{DS} = 0	I_{GSSF2}		1	mA dc
Gate current	3411	V_{GS} = -4 V dc, bias condition C, V_{DS} = 0	I_{GSSR2}			
2N7667UFBC					-800	µA dc
2N7668UFBC					-500	µA dc
2N7669UFBC					-300	µA dc
Drain current	3413	V_{GS} = 0, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}			
2N7667UFBC					1000	µA dc
2N7668UFBC					500	µA dc
2N7669UFBC					538	µA dc

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750 or industry standard		Symbol	Limits	Limits	Unit
	Method	Condition		Min	Max	
Subgroup 3 – Continued.						
Static drain to source on-state resistance	3421	V_{GS} = 5 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D1}$	$r_{DS(ON)2}$			
2N7667UFBC					0.018	Ω
2N7668UFBC					0.025	Ω
2N7669UFBC					0.050	Ω
Gate to source voltage (threshold)	3403	$V_{DS} = V_{GS}$, I_D = 3mA dc	$V_{GS(TH)2}$	0.7		V dc
Low temperature operation:		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} = V_{GS}$, I_D = 5 mA dc	$V_{GS(TH)3}$		3.0	V dc
Subgroup 4						
Gate charge	3471	Condition B, $I_D = I_{D1}$, V_{DD} = 50 percent rated V_{DS}				
Total gate charge			Q_G			
2N7667UFBC					11.4	nC
2N7668UFBC					11	nC
2N7669UFBC					7	nC
On gate to source charge			Q_{GS}			
2N7667UFBC					3.1	nC
2N7668UFBC					3.1	nC
2N7669UFBC					2.5	nC
On gate to drain charge			Q_{GD}			
2N7667UFBC					3.0	nC
2N7668UFBC					2.9	nC
2N7669UFBC					4.0	nC
Capacitance	3431	V_{DS} = 50 percent rated V_{DS}				
Input Capacitance			C_{ISS}			
2N7667UFBC					1300	pF
2N7668UFBC					1000	pF
2N7669UFBC					900	pF
Output Capacitance			C_{oss}			
2N7667UFBC					900	pF
2N7668UFBC					700	pF
2N7669UFBC					359	pF
Reverse Transfer Capacitance			C_{rss}			
2N7667UFBC					60	pF
2N7668UFBC					30	pF
2N7669UFBC					13	pF

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 5 (JANS).
Group E, subgroup 2.

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TABLE II. Group D inspection.

Inspection	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		Unit
1/ 2/ 3/	Method	Conditions		H Min	H Max	H Min	H Max	
Subgroup 1 Not applicable								
Subgroup 2		T_C = + 25°C						
Steady-state total dose irradiation (V_{GS} bias) 4/	1019	Condition A, V_{GS} = 5 V; V_{DS} = 0						
Steady-state total dose irradiation (V_{DS} bias) 4/	1019	Condition A, V_{GS} = 0; V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)						
Steady-state total dose irradiation (No bias) 4/	1019	V_{GS} = V_{DS} all shorted						
End-point electricals:								
Gate to source voltage (threshold)	3403	V_{DS} = V_{GS}, I_D = 5 mA dc	$V_{GS(TH)1}$	0.8	2.5	0.8	2.5	V dc
Gate current	3411	V_{GS} = +6 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		500		500	µA dc
Gate current	3411	V_{GS} = -4 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}					
2N7667UFBC					-400		-400	µA dc
2N7668UFBC					-250		-250	µA dc
2N7669UFBC					-150		-150	µA dc
Drain current	3413	V_{GS} = 0, bias condition C,	I_{DSS1}					
2N7667UFBC		V_{DS} = 100 percent of			400		400	µA dc
2N7668UFBC		rated V_{DS}			250		250	µA dc
2N7669UFBC					250		250	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 5 V dc, condition A, pulsed (see 4.5.1),	$r_{DS(ON)1}$					
2N7667UFBC		I_D = I_{D1}			0.01		0.01	Ω
2N7668UFBC					0.014		0.014	Ω
2N7669UFBC					0.030		0.030	Ω
Forward voltage	4011	V_{GS} = 0, condition A, pulsed (see 4.5.1), I_D = 0.5A	V_{SD}		3.0		3.0	V (pk)

1/ Also applies to UFBM suffix versions. For sampling plan see MIL-PRF-19500.
2/ Group D qualification may be performed prior to lot formation.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ Separate samples shall be pulled for each bias.

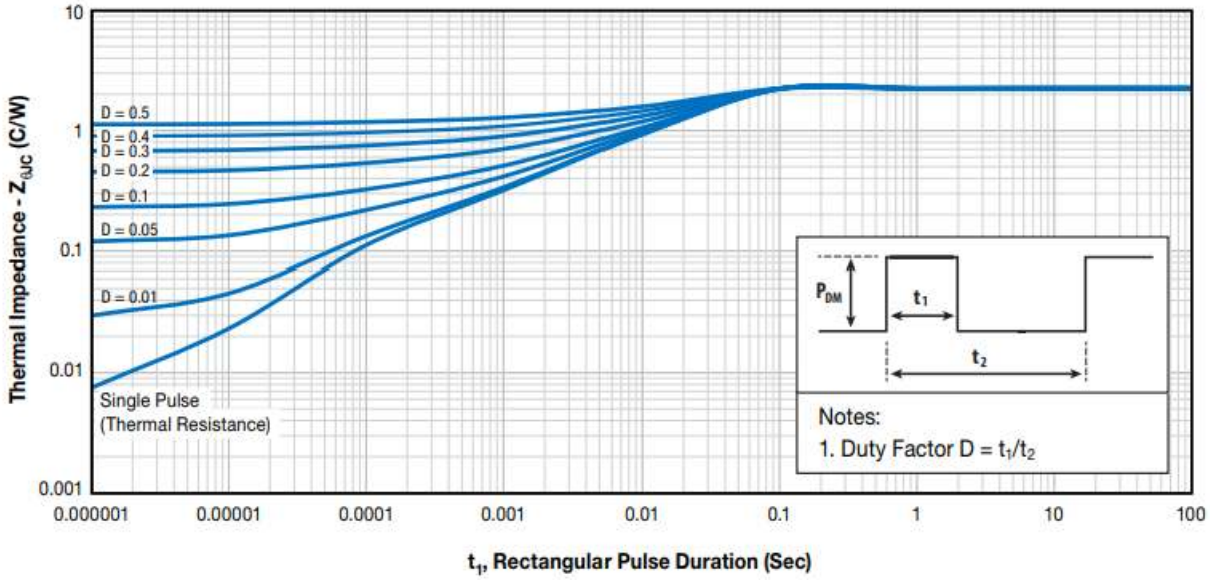
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TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

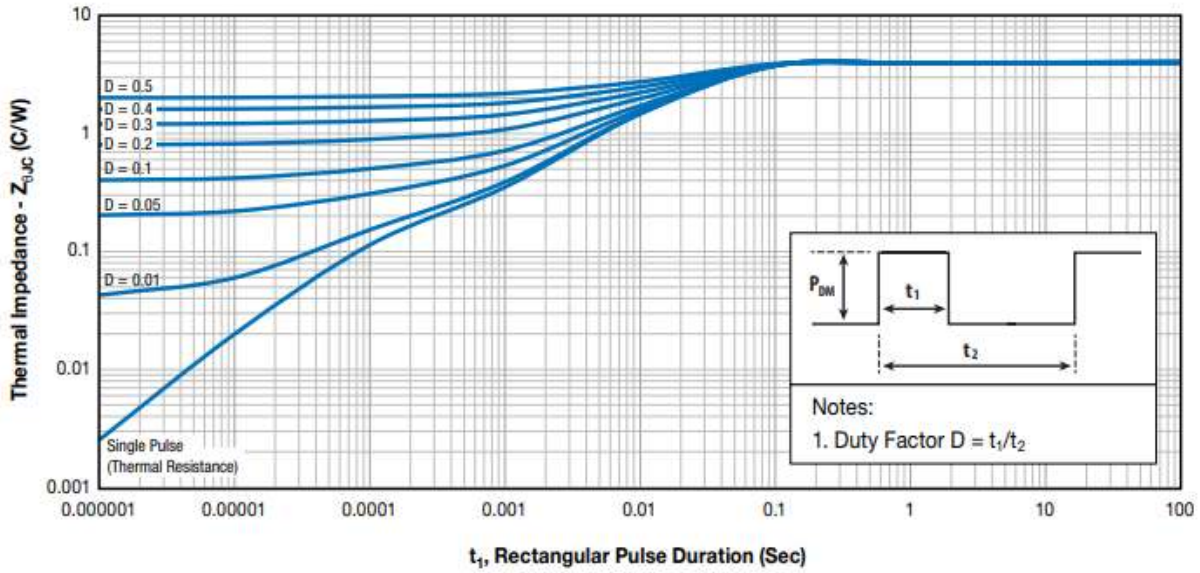
Inspection	MIL-STD-750 or industry standard		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Condition G, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2	
Subgroup 2 1/			45 devices c = 0
V_{DS} transient	JEP186	10K Transient pulses at 120% rated V_{DS} voltage. T_J = 125°C. This test may be performed at any time prior to lot formation.	
Steady-state gate bias 4/	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state reverse bias 4/	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			
User Guidelines for IR Thermal Imaging Determination of Die Temperature	JEP138	Thermal impedance curve development and/or characterization.	
Subgroup 7			3 devices c = 0
Resistance to soldering heat	2031	Condition (B or C) and I	
Subgroup 10			12 devices c = 0
Switching Dynamic Life Test 4/	JEP180	Test conditions shall be derived by the manufacturer. See 4.5.4.	
Electrical measurements		See table I, subgroup 2	
Subgroup 11			
SEE 2/ 3/	1080	See MIL-STD-750 method 1080, 4.5.5, and 6.2.	
Subgroup 12			4 devices
Surface Mount Device Package Integrity 5/	2039	Twist Bend temp cycle visual, Condition C, F, T (4 devices each condition)	

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE effect testing may be performed prior to lot formation.
3/ Device qualification to a higher level LET is sufficient to qualify all lower level LETs.
4/ For initial qualification perform 2000 hrs and then subsequent Group E testing perform 1000 hrs.
5/ For initial package qualification.

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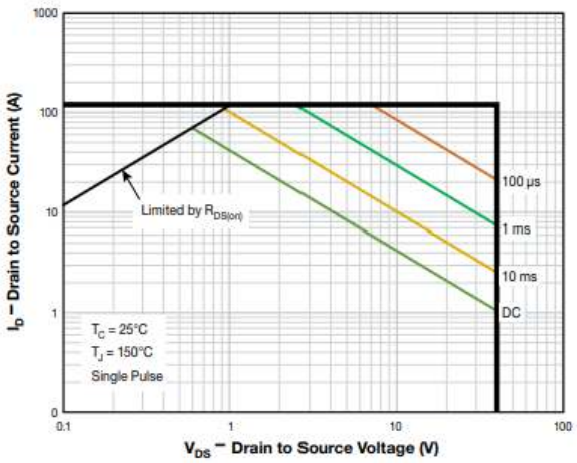
2N7667UFBC, 2N7668UFBC, 2N7667UFBM, 2N7668UFBM



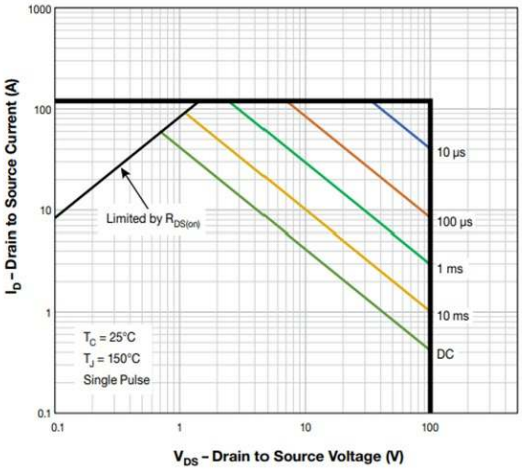
2N7669UFBC, 2N7669UFBM

FIGURE 4. Thermal impedance graph.

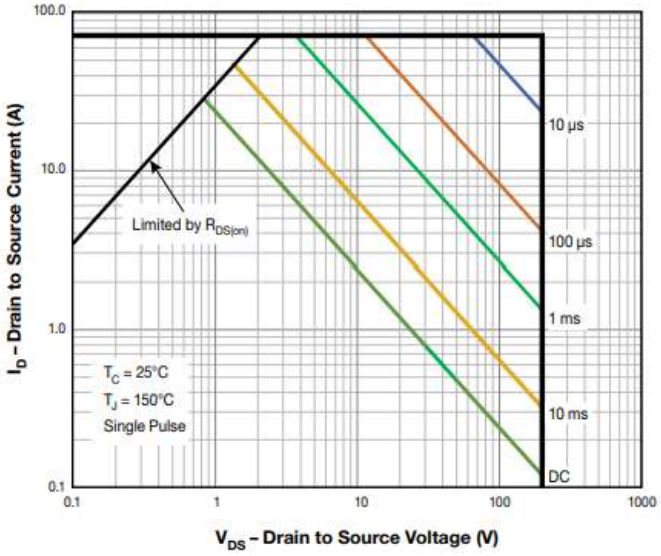
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2N7667UFBC, 2N7667UFBM



2N7668UFBC, 2N7668UFBM



2N7669UFBC, 2N7669UFBM

FIGURE 5. Safe-operating-area graph.

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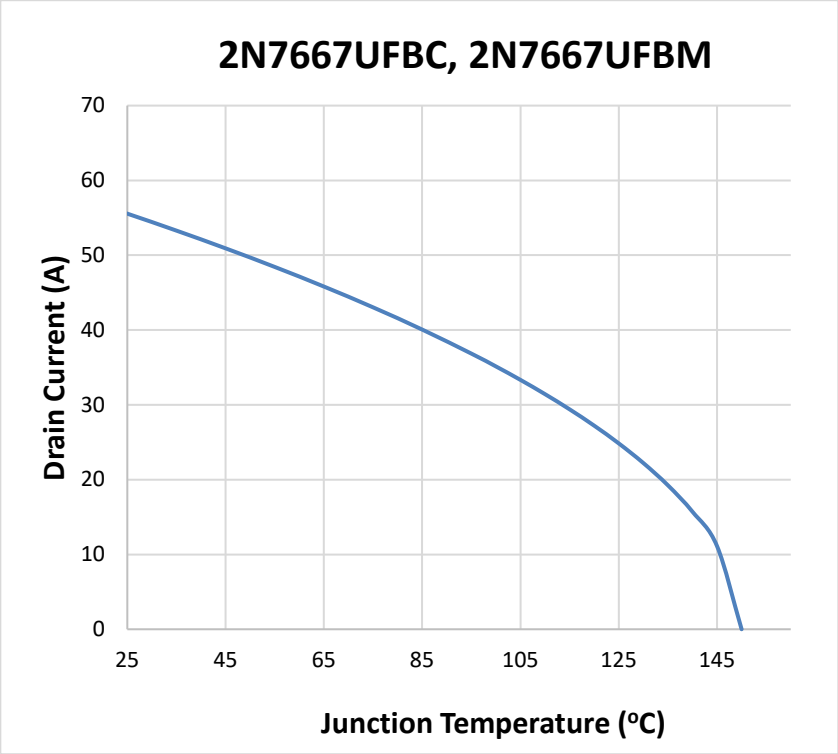


FIGURE 6. Drain Current vs Temperature graph.

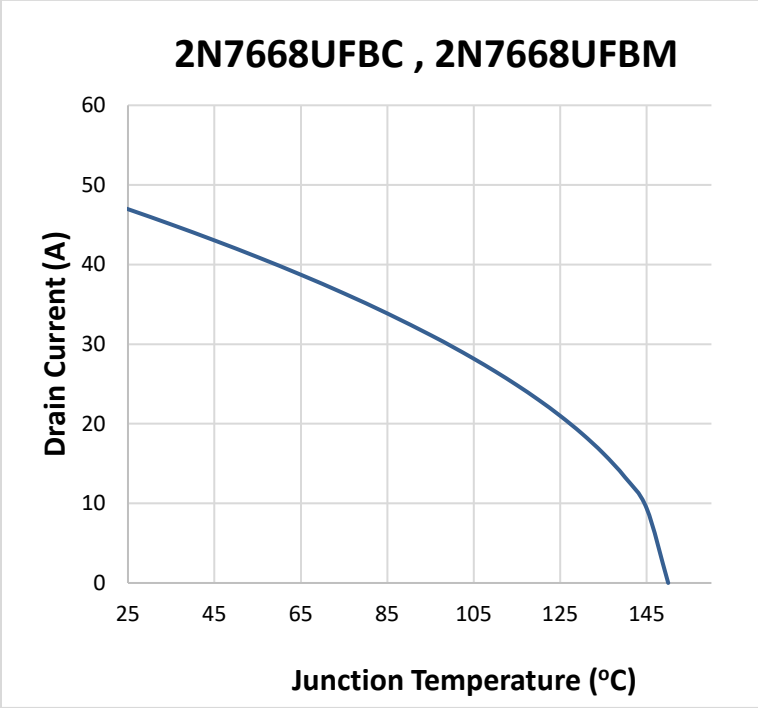


FIGURE 6. Drain Current vs Temperature graph - continued.

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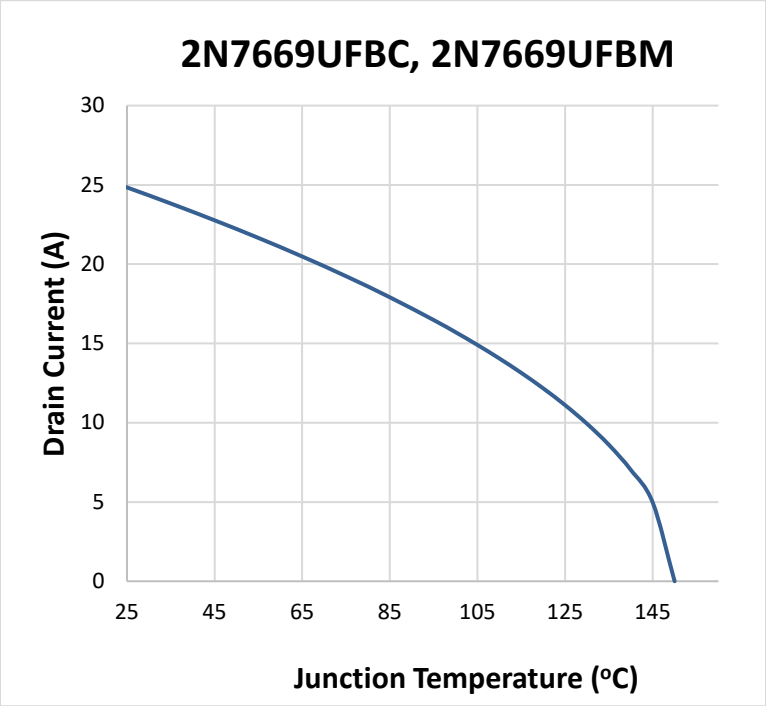


FIGURE 6. Drain Current vs Temperature graph - continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract or order.

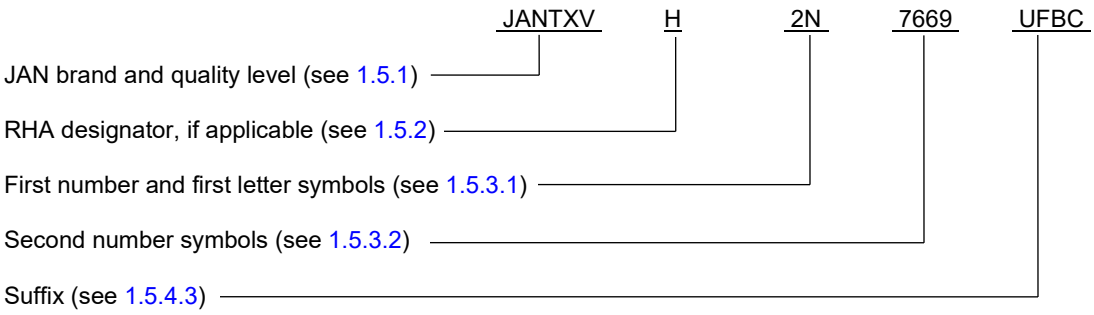
f. If specific SEE characterization conditions are desired (see section 6.7 and table IV), manufacturer's cage code should be specified in the contract or order.

g. If SEE testing data is desired, it should be specified in the contract or order.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.4 PIN construction example. The PINs for encapsulated devices are construction using the following form.



6.5 List of PINs.

6.5.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level with
JANTXVH2N7667UFBC	JANSH2N7667UFBC
JANTXVH2N7668UFBC	JANSH2N7668UFBC
JANTXVH2N7669UFBC	JANSH2N7669UFBC
JANTXVH2N7667UFBM	JANSH2N7667UFBM
JANTXVH2N7668UFBM	JANSH2N7668UFBM
JANTXVH2N6769UFBM	JANSH2N7669UFBM

1/ The M suffix represents a device with a metal lid.
2/ The PIN is also available without an RHA designator.

6.5.2 PINs for unencapsulated devices (die). The following is a list of possible PINs for unencapsulated devices available on this specification sheet.

PINs for devices of the "JANHC" quality level	PINs for devices of the "JANKC" quality level with
JANHCA2N7667	JANKCA2N7667
JANHCA2N7668	JANKCA2N7668
JANHCA2N6769	JANKCA2N7669

6.6 Cross-reference list. The following table shows the manufacture generic P/N and its associated military P/N.

Generic P/N	Military P/N	Package & Termination Configuration
FBG04N30B	2N7667UFBC	4-pad, Ceramic Lid
FBG10N30B	2N7668UFBC	4-pad, Ceramic Lid
FBG20N18B	2N7669UFBC	4-pad, Ceramic Lid

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6.7 Application data.

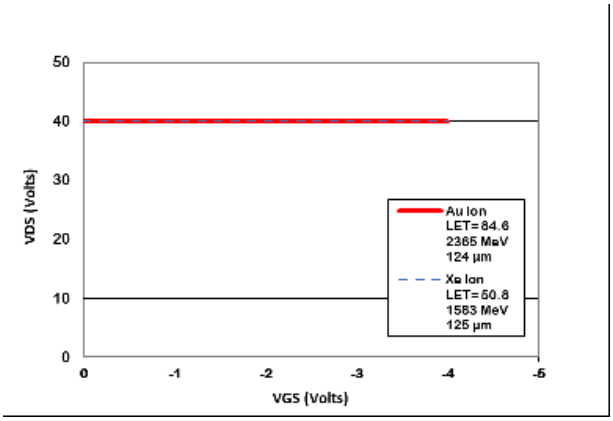
6.7.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer's characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.

TABLE IV. Manufacturers characterization conditions.

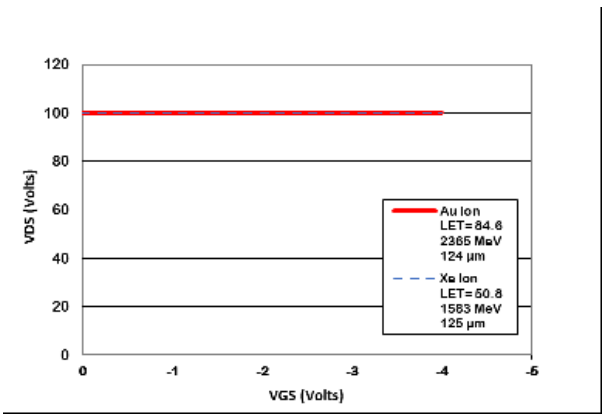
Manufactures cage	Inspection 1/	MIL-STD-750		Sample plan
		Method	Conditions	
8QKQ6	SEE 1/ 2/	1080	See MIL-STD-750E method 1080.1 dated 20 November 2006. See figure 7	
	Electrical measurements		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	3 devices
	SEE irradiation:		Fluence = 1E7 ions/cm^2 Flux = 2E3 to 2E4 ions/cm^2/sec, temperature = 25º ±5 °C Surface LET = 50 MeV-cm^2/mg (Si) ±5.0 %, range =131 µm (Si) ±7.5%, energy = 1653 MeV ±7.5% In-situ bias conditions: V_{DS} = 100 % rated, and V_{GS} = -4 V Surface LET = 84 MeV-cm^2/mg (Si) ±5 %, range = 130 µm (Si) ±7.5%, Energy = 2482 MeV ±10%	
	2N7667UFBC 2N7668UFBC 2N7669UFBC		In-situ bias conditions: VDS = 40 V and VGS = -4 V In-situ bias conditions: VDS = 100 V and VGS = -4 V In-situ bias conditions: VDS = 150 V and VGS = -4 V	
	Electrical measurements		I_{GSSF1}, I_{GSSR1}, and I_{DSS1}	

1/ Also applies to UFBM suffix versions.
2/ I_{GSSF1}, I_{GSSR1}, and I_{DSS1} were examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.

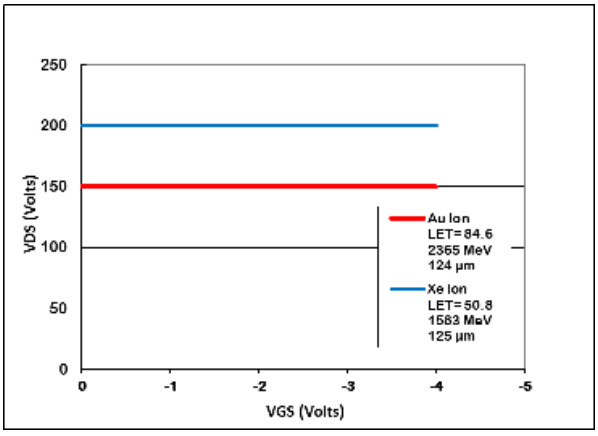
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2N7667UFBC



2N7668UFBC



2N7669UFBC

FIGURE 7. Cage 8QKQ6 typical SEE response graph.

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6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or DSN 850-6939.

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2023-032)

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